

MARGARET A LAWRIK

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RESEARCH INTERESTS

I apply big-data analytics, high-performance computing, and geostatistical methods to tackle complex issues. My interdisciplinary research encompasses climate change adaptation and resilience, sustainable development, and environmental justice. Collaborating with local communities, government agencies, and research institutions, I seek solutions to socio-environmental issues, emphasizing clear science communication and visualization.

EDUCATION

North Carolina State University (NCSU) Raleigh, NC
Ph.D. Geospatial Analytics February 2026

- Dissertation - Accounting for Regulatory Zoning in Land Change Science and Modeling
- **Relevant Coursework** - Remote Sensing of the Environment | Geosimulation and Computation | Geospatial Data Mining

North Carolina State University (NCSU) Raleigh, NC
B.S. Environmental Sciences May 2020

- Concentration - Geospatial Information Science (GIS)
- Minors - Computer Programming, Bassoon Performance
- GPA - 4.0/4.0 Degree Honors - Valedictorian, Summa cum laude, Dean's List (2016-2020)

PROFESSIONAL APPOINTMENTS

Mapping Analyst May 2025 - Present
Geovisualization and Communication | Southeast Climate Adaptation Science Center (SE CASC) Raleigh, NC

- Collaborate with interdisciplinary team to translate research products for non-technical audiences
- Develop interactive map-based project explorer and resource hub (Python, ArcGIS Pro, ArcGIS Online)

Graduate Research Assistant August 2021 - Present
Center for Geospatial Analytics | NCSU Raleigh, NC

- **Understanding the influence of planning policy on future land use** (Python, R, C, GRASS GIS)
 - Leveraging machine learning and high-performance computing for zoning classification using vector and remote sensing-derived data
 - Quantifying the historical influence of zoning using geostatistical methods and quasi-experimental design
 - Geospatial development of urban growth model to integrate land use planning realities
- **Modeling human dimensions of flooding** (Python, QGIS, GRASS GIS)
 - Assessed the unintended consequences of FEMA's 100-year floodplain across contiguous United States using remote sensing-derived data
 - Processed and visualized projected land change in response to sea-level rise and frequent flooding

Teaching Assistant Spring 2025 & Spring 2023
Fundamentals of Geospatial Information Science and Technology Raleigh, NC

- Class size: 80-100 students
- Held weekly office hours, advised students on developing geospatial workflows and visualizations (ArcGIS Pro, ArcGIS Online, ArcPy, QGIS)

Graduate Research Intern June - September 2024
Human Geography Group | Oak Ridge National Laboratory Oak Ridge, TN

- Assessed zoning's influence on human population dynamics and urban land use across 20 years in North Carolina using vector and remote sensing-derived data (Python, GRASS GIS)

Research Assistant September 2019 - June 2021
Center for Geospatial Analytics | NCSU Raleigh, NC

- **Food-Energy-Water-Transportation Systems Lab** | February 2021 - June 2021
 - Developed project monitoring web tools in collaboration with the North Carolina Department of Transportation (ArcGIS Pro, ArcPy, Python)

- Geospatial database development and management using PostgreSQL
- **Biological Invasions Lab** | May 2020 - February 2021
 - Geospatial development on Pest or Pathogen Spread (PoPS) model to implement eight additional dispersal kernels and add deterministic functionality (C++)
- **Urban Systems Lab** | September 2019 - May 2020
 - Geospatial development on FUTure Urban Regional Environment Simulation (FUTURES) to measure and reduce parameter uncertainty (C, GRASS GIS)

Undergraduate Research Assistant

May - August 2019

North Carolina Institute for Climate Studies

Asheville, NC

- Built a geospatial database of 10 datasets at four spatial scales over a 20-year period to analyze suicide rates in NC (SaTScan, ArcGIS Pro, R)

PUBLICATIONS

Peer Reviewed Articles

Sanchez, G.M., **Lawrimore, M.A.**, Petrasova, A., Vogler, J.B., Collins, E., Petras, V., Harper, T., Butzler, E., Meentemeyer, R.K. (2024). The Safe Development Paradox of the United States Regulatory Floodplain. PLoS ONE 19(12): e0311718.

Lawrimore, M. A., Sanchez, G. M., Cothron, C., Tulbure, M. G., BenDor, T. K., Meentemeyer, R. K. (2024). Creating spatially complete zoning maps using machine learning. Computers, Environment and Urban Systems, 112, 102157. <https://doi.org/10.1016/j.compenvurbsys.2024.102157>

Sanchez, G.M., Petrasova, A., Skrip, M., Collins, E., **Lawrimore, M.A.**, Vogler, J.B., Terando, A., Vukomanovic, J., Mitasova, H., Meentemeyer, R.K. (2023). Spatially interactive modeling of land change identifies location-specific adaptations most likely to lower future flood risk. Sci Rep 13, 18869. <https://doi.org/10.1038/s41598-023-46195-9>

Sugg, M.M., Woolard, S., **Lawrimore, M.A.**, Micheal, K.D., Runkle, J.D. (2021). Spatial Clustering of Suicides and Neighborhood Determinants in North Carolina, 2000 to 2017. Appl. Spatial Analysis 14, 395–413. <https://doi.org/10.1007/s12061-020-09364-1>

Under Review

Lawrimore, M.A., Sanchez, G.M., BenDor, T.K., Meentemeyer, R.K. Quantifying relationships between U.S. regulatory zoning and landscape-scale trends in impervious surface development. Under Review: Cities.

Articles In Preparation

Lawrimore, M.A., Sanchez, G.M., Petrasova, A., BenDor, T.K., Vukomanovic, J., Meentemeyer, R.K. Incorporating regulatory zoning in land change modeling improves predictive performance and planning relevance. In preparation for Landscape and Urban Planning

DATA AND SOFTWARE RELEASES

Lawrimore, M.A., Sanchez, G.M., Cothron, C., Tulbure, M.G., BenDor, T.K., Meentemeyer, R.K. (2023). Predicted Spatially Complete Zoning Map of North Carolina [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.8136886>

Petrasova, A., Sanchez, G.M., **Lawrimore, M.A.**, Vogler, J.B., Collins, E.L., Petras, V., Harper, T., Butzler, E., and Meentemeyer, R.K. (2023). FUTURES v2: Status Quo Projections of Future Patterns of Urbanization Across the Conterminous United States from 2020 to 2100: U.S. Geological Survey data release, <https://doi.org/10.5066/P94N3ICH>.

Petrasova, A., Sanchez, G.M., Skrip, M.M., Collins, E.L., **Lawrimore, M.A.**, Vogler, J.B., Terando, A., Vukomanovic, J., Mitasova, H., and Meentemeyer, R.K. (2023). FUTURES v3: Scenarios of Future Patterns of Urbanization in Response to Sea Level Rise and Frequent Flooding Across the Southeast United States from 2020 to 2100: U.S. Geological Survey data release, <https://doi.org/10.5066/P9BD5V4B>.

GRANTS AND FELLOWSHIPS

PI: Sanchez, G.M.; **I: Lawrimore, M.A.** (Awarded) Amount: \$10,000. Smart Zoning for Coastal Flood Adaptation and Resilience. NC Sea Grant program. September 2022 - September 2023.

Lawrimore, M.A. (Awarded) Amount: \$4,000. University Graduate Fellowship (2021-2022). North Carolina State University.

PRESENTATIONS

Invited Talks

Lawrimore, M.A., et al. (November 2025). The Unintended Consequences of the 100-Year Floodplain Across the US Southeast. Coffee with a Climate Expert: Session 3, Flooding and Development. Southern Environmental Law Center. Virtual *oral presentation*.

Lawrimore, M.A., et al. (April 2025). When Flood Management Policies Miss the Mark: Unintended Consequences of the 100-Year Floodplain. Leveraging Geospatial Tools for Effective Environmental Planning Webinar, Southeast Climate Adaptation Science Center. Virtual *oral presentation*.

Lawrimore, M.A., et al. (September 2024). Leveraging Machine Learning to Fill Zoning Data Gaps. Smart Cities Consortium. University of Texas at Austin. Virtual *oral presentation*.

Lawrimore, M.A., et al. (February 2024). Creating Spatially Complete Zoning Maps Using Machine Learning. NCSU Geospatial Forum. Raleigh, NC. *Oral presentation*.

Conference Presentations

Lawrimore, M.A., et al. (April 2024). Creating Spatially Complete Zoning Maps Using Machine Learning. International Association for Landscape Ecology - North America. Oklahoma City, OK. *Oral presentation*.

Lawrimore, M.A., et al. (December 2023). Spatially Interactive Modeling of Urban Growth and Human Migration Driven by Future Flood Hazard Conditions. AGU23. San Fransico, CA. *Oral presentation*.

Lawrimore, M.A., et al. (October 2023). Forecasting Human Mobility and Development Patterns Driven by Future Flood Hazard Conditions. Envisioning Urban Futures Symposium. NCSU, Raleigh, NC. *Poster presentation*.

Lawrimore, M.A., et al. (March 2023). Creating Spatially Continuous Zoning Maps Using Machine Learning. CNR Graduate Research Symposium. NCSU, Raleigh, NC. *Poster presentation*.

Lawrimore, M.A., et al. (November 2022). Smart Zoning for Coastal Flood Adaptation and Resilience. North Carolina Coastal Conference. Raleigh, NC. *Poster presentation*.

Lawrimore, M.A., et al. (September 2022). Forecasting Scenarios of Human Mobility and Shifts in Development Patterns Driven by Future Flood Hazard Conditions. 2022 Southeast Climate Adaptation Science Symposium. Gulf Shores, AL. *Poster presentation*.

RESEARCH ENGAGEMENT

Research Partnership and Collaboration

Understanding Zoning's Influence on Human Population

June 2024 - December 2024

- Partner: Human Geography Group | Oak Ridge National Laboratory
- Goal: Assess the influence of zoning regulations on human land use and population dynamics

Smart Zoning for Coastal Flood Adaptation and Resilience

September 2022-September 2023

- Stakeholder: Town of Leland, NC, Economic and Community Development Department
- Partner: North Carolina Sea Grant
- Goal: Assess zoning regulations, flood exposure, and future development to identify areas for strategic interventions
- Summary Report: Lawrimore, M.A., Sanchez, G.M. (2023). Smart zoning for coastal flood resilience and adaptation. Submitted to the Community Development Planning Department at the Town of Leland, NC.

Civic Technology Volunteer

Code with the Carolinas and the National Zoning Atlas

March 2023 - January 2025

- Co-lead bi-weekly meetings to introduce volunteers to civic-technology volunteering
- Assist the organization in gathering zoning data for North and South Carolina and contribute to the National Zoning Atlas

AWARDS

- NCSU Graduate Student Association Conference Travel Award (Spring 2025). Amount: \$500.
- North Carolina State University Center for Geospatial Analytics Travel Award (Spring 2025). Amount: \$500.
- North Carolina State University Center for Geospatial Analytics Travel Award (Fall 2024). Amount: \$1000.
- IALE-NA Student Travel Award (Spring 2024) Amount: \$500.
- North Carolina State University Center for Geospatial Analytics Travel Award (Fall 2023). Amount: \$800.
- First Prize - Outreach. Predicting Municipal Zoning in Wake County, NC. 2022. NCSU Graduate Student Association Infographic Contest.
- Honorable mention. Flood-prone Development in Charleston, SC. 2022. NCSU Research Image Contest. Work showcased across University promotional materials, websites, and social media 20+times from 2022-2025.

LEADERSHIP

Co-President

May 2023 - May 2024

Geospatial Graduate Student Organization | NCSU

Elected March 2023

- Organize and lead events, organization spokesperson, student advocate, student-administration liaison

Mentorship

August 2021 - May 2022

Center for Geospatial Analytics | NCSU

- Mentored undergraduate research assistant on lab protocols, data and time management, and how to transition from undergraduate studies to the workforce

TECHNICAL EXPERTISE

• **Programming Languages:**

Advanced Proficiency: Python, R

Experience: C++, C, Java, Bash, HTML

Geospatial Libraries

Python: geopandas, rasterio, scikit-learn | R: terra, sf

• **High Performance Computing:** NCSU Hazel Cluster

Libraries for parallel processing: dask, xarray, joblib

• **Geographic Information Systems (GIS):**

ArcGIS Pro, ArcGIS Online, GRASS GIS, QGIS

• **Database Management:** PostgreSQL

• **Cloud Computing:** Google Earth Engine

• **Co-production:** Git, GitHub